

ZHONGSHAN, Antarctica

WMO: 895730

Lat: 69.374S

Lon: 76.372E

Elev: 59

StdP: 14.66

Time Zone: 5.00 (E05)

Period: 91-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -29.8	(c) -23.8	(d) -37.5	(e) 0.6	(f) -26.3	(g) -33.9	(h) 0.8	(i) -20.9	(j) 68.9	(k) 26.6	(l) 61.3	(m) 14.8	(n) 4.4	(o) 250	(p) 1.244

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 6.2	(c) 40.1	(d) 32.8	(e) 38.6	(f) 31.9	(g) 37.0	(h) 30.9	(i) 33.6	(j) 38.5	(k) 32.5	(l) 37.2	(m) 31.5	(n) 36.1	(o) 8.8	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
30.0	24.1	33.2	27.9	21.9	32.8	26.1	20.2	32.6	12.5	38.8	12.1	37.6	11.7	36.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 53.4	(b) 45.0	(c) 38.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 53.4	(b) 45.0	(c) 38.9	(e) -34.0	(f) 35.2	(g) 6.7	(h) 1.9	(i) -38.8	(j) 36.6	(k) -42.7	(l) 37.7	(m) -46.5	(n) 38.8	(o) -51.3	(p) 40.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	32.7	26.4	16.2	7.2	3.1	4.7	0.3	2.6	3.3	9.8	23.1	31.8
	DBStd	14.41	2.83	4.28	6.61	8.64	10.46	10.19	13.26	10.99	10.90	7.54	6.17	3.51
	HDD50	13376	537	660	1048	1284	1455	1359	1540	1470	1402	1248	808	565
	HDD65	18851	1002	1080	1513	1734	1920	1809	2005	1935	1852	1713	1258	1030
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.6	11.8	17.6	19.0	16.5	17.9	19.2	17.7	19.0	16.7	16.9	14.2	13.2
Precipitation	PrecAvg	13.2	1.5	1.5	1.3	0.8	0.9	0.9	1.1	0.9	0.8	0.9	1.3	1.2
	PrecMax	17.7	4.2	2.8	2.7	2.1	2.1	2.2	2.9	3.1	2.1	2.0	3.3	2.4
	PrecMin	6.7	0.5	0.7	0.4	0.3	0.3	0.2	0.0	0.1	0.0	0.2	0.5	0.4
	PrecStd	2.0	0.7	0.6	0.6	0.4	0.4	0.4	0.6	0.6	0.4	0.4	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	38.9	30.7	26.7	27.7	27.0	26.9	27.5	25.8	28.6	37.4	42.7
		MCWB	34.3	32.1	26.3	23.3	23.3	22.8	24.5	22.9	22.6	23.5	29.9	34.8
	2%	DB	40.1	35.1	28.7	23.9	24.1	23.4	24.2	22.3	23.9	25.8	35.1	40.4
		MCWB	32.9	29.9	24.7	21.4	21.2	20.2	22.4	19.2	20.5	21.9	29.1	33.0
	5%	DB	38.7	33.8	26.8	21.9	21.5	21.3	21.2	19.7	20.4	23.6	33.3	38.9
		MCWB	32.0	28.9	23.6	19.2	19.1	18.6	19.4	17.0	17.3	20.2	28.1	32.0
	10%	DB	37.3	32.3	24.9	19.0	17.5	18.8	18.0	17.2	17.6	21.0	31.7	37.4
		MCWB	31.3	27.7	21.8	16.6	14.7	16.4	16.0	15.0	15.0	17.8	26.9	31.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.3	33.2	27.3	24.1	23.9	23.7	25.2	22.9	23.6	25.3	31.3	35.4
		MCDB	39.6	36.2	29.6	26.0	26.9	26.7	26.5	26.6	25.2	27.1	35.1	41.9
	2%	WB	34.0	31.2	25.4	21.6	21.4	20.7	22.6	19.6	20.5	22.4	29.7	33.4
		MCDB	38.6	33.9	27.9	23.9	23.4	22.7	23.9	21.8	23.6	25.4	33.9	38.9
	5%	WB	32.9	29.7	23.9	19.1	19.0	18.8	19.7	17.3	17.5	20.2	28.7	32.4
		MCDB	37.4	32.6	26.6	21.2	21.6	21.2	21.4	19.6	20.2	23.4	32.9	37.7
	10%	WB	31.8	28.4	21.9	16.7	15.0	16.4	16.0	15.0	15.0	18.0	27.3	31.5
		MCDB	36.3	31.8	24.8	19.1	17.4	18.8	18.0	17.3	17.4	21.0	31.6	36.6
Mean Daily Temperature Range	5% DB	MDBR	6.2	5.8	6.2	7.4	8.2	8.4	8.8	9.6	8.3	8.6	8.8	7.7
		MCDBR	8.3	7.3	6.5	8.3	8.9	8.7	7.1	9.5	6.7	9.0	9.1	9.5
		MCWBR	5.9	6.2	5.9	8.2	8.8	8.3	6.8	8.7	6.0	8.0	7.2	6.8
	5% WB	MCDBR	7.4	6.0	6.1	8.8	8.5	9.6	7.0	10.0	8.0	9.3	7.8	8.4
		MCWBR	5.6	5.7	6.0	9.0	8.5	9.4	6.8	9.5	7.5	8.7	6.5	6.5
Clear-Sky Solar Irradiance	taub	0.257	0.246	0.219	0.163	0.121	N/A	0.124	0.158	0.202	0.217	0.242	0.258	
	taud	2.317	2.347	2.308	2.104	1.904	N/A	1.884	2.061	2.222	2.236	2.247	2.264	
	Ebn at Noon	315	299	272	206	29	0	19	202	274	308	318	319	
	Edh at Noon	34	28	21	11	1	0	1	11	22	30	36	37	
All-Sky Solar Radiation	RadAvg	2547	1655	811	237	19	0	3	119	592	1430	2351	2876	
	RadStd	93	79	38	11	1	0	0	7	31	52	81	102	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page